TIETZE TRANSFORMATIONS PRESERVE PRESENTED SETS AND ARE COMPLETE UNDER CHOICE

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Editorial note. The completeness direction retains the original axiom of choice and the author disclosure that §2.3 has not yet been fully formalized in Agda; collection does not add a formalization.

Allowed standard prerequisites (editorial).

- Homotopy type theory with univalence, higher inductive coequalizers and propositional/set truncation. Full author §1 and exactsource references retained.
- Axiom of choice for the completeness direction, in the exact formulation of author Axiom18. Author §2.3 explicitly states this assumption and that the section is not yet fully formalized in Agda.

1 1-polygraphs

Before introducing 1-polygraphs, we first need to present the (quite trivial) 0-dimensional variant. A 0-polygraph P is simply a type whose elements are called 0-generators. A sphere (x, y) in a 0-polygraph P is a pair of elements of P, and we write $\mathrm{S}\,P$ for the type of such objects. If we write $\mathrm{S}^0 = \{0, 1\}$ for the type with two elements, corresponding to the canonical 0-sphere, it can be noted that a sphere in a 0-polygraph P corresponds to an arrow $\mathrm{S}^0 \to P$. Given a 0-polygraph P, we call $\|P\|_{-1}$ the proposition presented by P.

1.1 Internal and external 1-polygraphs

1-polygraphs.

A 1-polygraph P consists of a 0-polygraph $\mathrm{U}\,P$ (the underlying 0-polygraph of P) together with a function $\mathrm{S}\,\mathrm{U}\,P \to \mathcal{U}$, which to any sphere (x, y) in the underlying 0-polygraph associates a type $P(x, y)$ whose elements are called the 1-generators of the 1-polygraph. In other words, 1-polygraphs correspond to type-theoretic directed graphs. A 1-polygraph is discrete when all the types $\mathrm{U}\,P$ and $P(x, y)$ are sets. In the following, a sphere in $\mathrm{U}\,P$ is called a 0-sphere of P.

1-generators.

The elements of $\mathrm{U}\,P$ are called 0-generators or 0-cells of P. We sometimes write $\mathrm{S}_0\,P := \mathrm{S}\,\mathrm{U}\,P$ for the 0-spheres, as well as $P_0 := \mathrm{U}\,P$ and $P_1 := \Sigma((x, y) : \mathrm{S}_0\,P).P(x, y)$ for the type of 0- and 1-generators of P. We also write $\partial^-, \partial^+ : P_1 \to P_0$ for the source and target functions (defined by $\partial^- := \pi \circ \pi$ and $\partial^+ := \pi' \circ \pi$).

1-cells.

Given a 1-polygraph P and a 0-sphere (x, y), we write $P^*(x, y)$ for the type of 1-cells in P from x to y. This family of types is defined inductively as the smallest one such that

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- for every 0-cell $x : \mathrm{U}\,P$, we have a 1-cell $\mathrm{id}_x : P^*(x, x)$,
- for every 0-cells $x, y, z : \mathrm{U}\,P$, 1-generator $a : P(x, y)$ and 1-cell $u : P^*(y, z)$, we have a 1-cell $a :: u : P^*(x, z)$,
- for every 0-cells $x, y, z : \mathrm{U}\,P$, 1-generator $a : P(y, x)$ and 1-cell $u : P^*(y, z)$, we have a 1-cell $a ::' u : P^*(x, z)$.

Informally, cells in $P^*(x, y)$ are undirected paths from x to y in the graph corresponding to P (where the two last cases respectively correspond to taking an arrow a forward and backward). In the following, we often write $a : x \to y$ for a 1-generator $a : P(x, y)$ and $u : x \overset{*}{\to} y$ for a 1-cell $u : P^*(x, y)$.

Composition.

Given two 1-cells $u : x \overset{*}{\to} y$ and $v : y \overset{*}{\to} z$, we define their composite $u \cdot v : x \overset{*}{\to} z$ by induction on u by

$\mathrm{id} \cdot v := v$ $(a :: u) \cdot v := a :: (u \cdot v)$ $(a ::' u) \cdot v := a ::' (u \cdot v)$

Lemma 1 ([13, comp0-assoc]).

Composition is associative and unital.

Given a 1-cell $u : x \overset{*}{\to} y$, we define its opposite $u^{\mathrm{op}} : y \overset{*}{\to} x$ by induction on u by

$\mathrm{id}^{\mathrm{op}} := \mathrm{id}$ $(a :: u)^{\mathrm{op}} := a \mathbin{\overline{::}}' u^{\mathrm{op}}$ $(a ::' u)^{\mathrm{op}} := a \mathbin{\overline{::}} u^{\mathrm{op}}$ where $a \mathbin{\overline{::}} u$, which corresponds to appending a to u, is defined by

$a \mathbin{\overline{::}} \mathrm{id} := a :: \mathrm{id}$ $a \mathbin{\overline{::}} (b :: u) := b :: (a \mathbin{\overline{::}} u)$ $a \mathbin{\overline{::}} (b ::' u) := b ::' (a \mathbin{\overline{::}} u)$ and $a \mathbin{\overline{::}}' u$, corresponding to appending a backward to u, is defined similarly.

Spheres.

A sphere s in a 1-polygraph P consists of a 0-sphere (x, y) together with a pair of 1-cells $f, g : x \overset{*}{\to} y$. The underlying 0-sphere (x, y) is noted $\mathrm{U}\,s$. The cell f (resp. g) is called the source (resp. target) of s and noted $\partial^- s$ (resp. $\partial^+ s$), so that the sphere s is the triple $(\mathrm{U}\,s, \partial^- s, \partial^+ s)$. We write $\mathrm{S}\,P$ for type of spheres in a 1-polygraph.

Coherent 1-cells.

In the definition of 1-cell, we think of $a :: -$ and $a ::' -$ as respectively pre-composing by the 1-cell a and its inverse. However, nothing enforces that they are actually inverse operations, i.e. that we have

$a ::' a :: u = u$ $a :: a ::' u = u$ for any 1-generator a and 1-cell u of suitable types. We define the types of coherent 1-cells $P^{\sim}(x, y)$, indexed by 0-spheres (x, y), as the quotient of the types of 1-cells $P^*(x, y)$ under the congruence generated by the above relations. This type is the "right" definition for the 1-cells of P, in the sense that it gives a description of the free groupoid generated by P [9]. However, non coherent 1-cells provides representatives for those which are often simpler to work with. Namely, by definition of coherent 1-cells as a quotient of 1-cells, we have

Lemma 2.

For $x, y : \mathrm{U}\,P$, the canonical map $f : P^*(x, y) \to P^{\sim}(x, y)$ is a surjection, i.e. the type $\| \mathrm{hfib}_f(u) \|_{-1}$ is inhabited for any $u : P^{\sim}(x, y)$.

1.2 Geometric realization

We introduce the "geometric realization" of a 1-polygraph P, which can be thought of as a type-theoretical graph with P_0 as vertices and P_1 as edges. The set presented by P is then defined as the type of connected components of this graph.

Geometric realization.

Given a 1-polygraph P its geometric realization $\overline{P}$ is the higher inductive type generated by the two constructors

$\iota : \mathrm{U}\,P \to \overline{P}$ $\gamma : ((x, y) : \mathrm{S}_0\,P) \to P(x, y) \to \iota x = \iota y$ More abstractly, it corresponds to the pushout on the left or to the coequalizer on the right

$$\begin{array}{ccc} P_1 \times \mathrm{S}^0 & \xrightarrow{f} & P_0 \\ {\scriptstyle \pi'}\downarrow & \ulcorner & \downarrow \\ P_1 & \longrightarrow & \overline{P} \end{array} \qquad\qquad P_1 \underset{\partial^+}{\overset{\partial^-}{\rightrightarrows}} P_0 \longrightarrow \overline{P}$$

where the horizontal arrow f sends an edge $a : x \to y$ and 0 (resp. 1) to x (resp. y), which corresponds to attaching 1-cells to a 0-dimensional cell complex. The associated induction principle is the following.

Proposition 3.

Suppose given an arbitrary type A. Any two functions $f_0 : P_0 \to A$ and $f_1 : (((x, y), a) : P_1) \to f_0(x) = f_0(y)$ induce a unique function $f : \overline{P} \to A$ such that $f \circ \iota = f_0$ and $f \circ \gamma = f_1$. We also have the expected dependent generalization.

By partial evaluation, the map γ induces, for $(x, y) : \mathrm{S}_0\,P$, a map $\gamma_{x,y} : P(x, y) \to \iota x = \iota y$, which extends as a map on cells $\gamma^*_{x,y} : P^*(x, y) \to \iota x = \iota y$ defined by induction by

$\gamma^*_{x,x}(\mathrm{id}) := \mathrm{refl}$ $\gamma^*_{z,y}(a :: u) := \gamma a \cdot \gamma^*_{x,y}(u)$ $\gamma^*_{z,y}(a ::' u) := \gamma a^- \cdot \gamma^*_{x,y}(u)$ Moreover, since this definition is compatible with the relations for coherent 1-cells, for every sphere $(x, y) : \mathrm{S}_0\,P$ we have the following.

Lemma 4.

The map $\gamma^*_{x,y}$ induces a map $\gamma^\sim_{x,y} : P^\sim(x, y) \to \iota x = \iota y$.

This map is an equivalence, so it provides a characterization of the paths in geometric realizations of graphs [13, KvR/1Polygraph/PathPresentation.agda]:

Theorem 5.

The map $\gamma^\sim_{x,y} : P^\sim(x, y) \to \iota x = \iota y$ is an equivalence.

Proof.

A map $\iota x = \iota y \to P^\sim(x, y)$ can be defined using the elimination principle for paths in $\overline{P}$ given in [9, Theorem 3], and both can be shown to be mutually inverse by induction. ◀

Presented set.

The set presented by a 1-polygraph P is the set of connected components of its geometric realization $\|\overline{P}\|_0$. This notion is to be understood as an analogue of a group presentation with generators and relations, but shifted down by one dimension: here, the 0-cells of P are the generators of the presented set, and the 1-cells are the relations.

Example 6.

Consider the polygraph P with $\mathrm{U}\,P = \mathbb{Z}$ and one relation $a_n : n \to n + 2$ for $n \in \mathbb{Z}$. The presented set is $\mathbb{Z}_2$.

We now prove some technical lemmas that will be useful later on.

Lemma 7.

For $x, y : \mathrm{U}\,P$ and $p : |\iota(x)|_0 = |\iota(y)|_0$ in $\|\overline{P}\|_0$, there merely exists a 1-cell $x \xrightarrow{*} y$ in P.

Proof.

By [19, Theorem 7.3.12], the type $|\iota(x)|_0 = |\iota(y)|_0$ coincides with $\|\iota(x) = \iota(y)\|_{-1}$. Since we are eliminating to a proposition, we can suppose given a path $p : \iota(x) = \iota(y)$ which by Theorem 5 induces a coherent cell $\tilde{p} : P^\sim(x, y)$, and we conclude by Lemma 2. ◀ The following shows that we can pick a representative 0-cell in every connected component [13, has-repr]:

Proposition 8.

The type $\Pi(x : \|\overline{P}\|_0).\|\Sigma(\hat{x} : \mathrm{U}\,P).|\iota(\hat{x})|_0 = x\|_{-1}$ is inhabited.

Proof.

Any proposition being a set [19, Lemma 3.3.4], it is enough to construct a map $f : \Pi(x : \overline{P}).\|\Sigma(\hat{x} : \mathrm{U}\,P).\iota(\hat{x}) = x\|_{-1}$ which can be done by using the elimination principle of $\overline{P}$ (Proposition 3). Namely, we define, for $x : \mathrm{U}\,P$, $f(\iota(x)) := |(x, \mathrm{refl}_{\iota(x)})|_{-1}$ and, for $x, y : \mathrm{U}\,P$ and $a : P(x, y)$, the path associated to it by f is canonically defined by the fact that we are eliminating to a proposition. ◀

2 Tietze transformations

Tietze transformations originate from the work of Tietze [18] who defined two transformations on group presentations:

(T0)

adding a definable generator,

(T1)

adding a derivable relation.

The equivalence relation they induce, the Tietze equivalence, characterizes when two presentations give rise to the same group, up to isomorphism. These transformations have then been generalized to a variety of contexts, including the very general one of polygraphs [1, 5] for presentations of (higher) categories, and have as well been shown to provide a interesting notion of weak equivalence [7]. Their fundamental property can be decomposed in two parts:

■

correctness: any two presentations which are Tietze equivalent present the same structure,

■

completeness: any two presentations of the same structure are Tietze equivalent.

Here, we think of a 1-polygraph P as presenting the set $\|\overline{P}\|_0$, and we define elementary transformations as explained above. We then show that they are correct and complete with respect to the presented set.

2.1 Definition

We define the two following families of Tietze transformations, which transform a given 1-polygraph P into another 1-polygraph, by attaching cells to it:

(T0)

Add a definable 0-generator: given a type X and a function $\partial : X \to \mathrm{U}\,P$, we construct the polygraph $P{\uparrow_0}\partial$ defined, for $(x, y) : \mathrm{S}_0\,P$, by

$$\mathrm{U}(P{\uparrow_0}\partial) := \mathrm{U}\,P \sqcup X \quad (P{\uparrow_0}\partial)(x, y) := \begin{cases} P(x, y) & \text{if } x : \mathrm{U}\,P \text{ and } y : \mathrm{U}\,P \\ x = \partial(y) & \text{if } x : \mathrm{U}\,P \text{ and } y : X \\ \bot & \text{otherwise} \end{cases} \quad \text{(T1)}$$

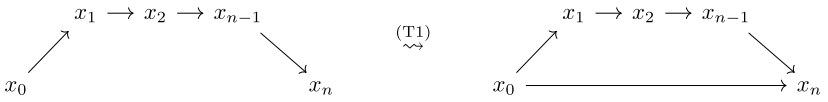
Add a derivable relation: given a function $\partial : \mathrm{S}_0\,P \to \mathcal{U}$ and a family of functions $\partial_{x,y} : \partial(x, y) \to (x \overset{*}{\to} y)$ indexed by 0-spheres $(x, y) : \mathrm{S}_0\,P$, we construct the polygraph $P{\uparrow_1}\partial$ defined, for $(x, y) : \mathrm{S}_0\,P$, by

$\mathrm{U}(P{\uparrow_1}\partial) := \mathrm{U}\,P$ $(P{\uparrow_1}\partial)(x, y) := P(x, y) \sqcup \partial(x, y)$

A transformation (T0) consists in adding, for every element $x : X$, a new 0-generator x to the polygraph together with a 1-generator $\tau_x : \partial x \to x$:

$$y \qquad \overset{\text{(T0)}}{\rightsquigarrow} \qquad y \xrightarrow{\tau_x} x$$

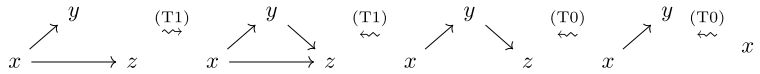
with $y \overset{\text{def}}{=} \partial x$, and τ_x is a notation for the element $\mathrm{refl}_x : (P{\uparrow_0}\partial)(y, x)$. A transformation (T1) consists in adding, for every $x, y : X$ and $a : \partial(x, y)$, a 1-generator $a : x \to y$:



It can be observed that we have a new hole after performing a transformation (T1), which justifies the importance of the set-truncation in the definition of the presented set. Note that the functions $\partial_{x,y}$ are not used in order to define Q in (T1), but they will be necessary to show the correctness of the transformation below. For instance, if we did not require those, we could make any polygraph coherent by taking $\partial(x, y) = \top$ for any $(x, y) : \mathrm{S}_0\, P$.

▶ **Remark 9.**
We could also consider a weaker notion of transformation (T1) where we only require the existence of the 1-cell associated to an added 1-generator, i.e. type the function as $\partial_{x,y} : \partial(x, y) \to \|x \xrightarrow{*} y\|_{-1}$. This variant would in fact be enough for the subsequent developments here.

Example 10.
We have the following series of Tietze transformations:



Under the Grothendieck duality between type families and fibrations [19, Section 4.8], the Tietze transformations (T1) can reformulated as follows (a similar reformulation for (T0) could also be given):

Proposition 11.
Transformations (T1) are equivalent to the following:
(T1')
Given a type $X : \mathcal{U}$ and function $\partial : X \to \Sigma((x, y) : \mathrm{S}_0\, P).P^*(x, y)$, we construct the polygraph $P{\uparrow_1}\partial$ with $\mathrm{U}(P{\uparrow_1}\partial) := \mathrm{U}\, P$ and, for $(x, y) : \mathrm{S}\, P$,

$$(P{\uparrow_1}\partial)(x, y) := P(x, y) \sqcup \mathrm{hfib}_{\pi\circ\partial}(x, y)$$

We define the Tietze equivalence as the equivalence relation generated by (T0) and (T1).

Transformations for 0-polygraphs.
The notion of Tietze transformation generalizes to polygraphs in any dimension. In particular, for 0-polygraphs, it can be formulated as follows. An elementary Tietze transformation of 0-polygraphs consists in
(T0)
Given a 0-polygraph P, a type X and a function $\partial : X \to P$, we consider the 0-polygraph $P{\uparrow_0}\partial := P \sqcup X$.

Two 0-polygraphs are Tietze equivalent when they are related by the smallest equivalence relation identifying P with $P{\uparrow_0}\partial$ for any P and ∂.

Proposition 12.
There exists a Tietze equivalence between two 0-polygraphs P and Q if and only if they present the same proposition, i.e. $\|P\|_{-1} = \|Q\|_{-1}$.

2.2 Correctness

Our aim is now to show the correctness property for Tietze transformations of 1-polygraphs: equivalent 1-polygraphs present the same set.

Theorem 13 (Correctness).
Given two Tietze equivalent 1-polygraphs P and Q, we have $\|\overline{P}\|_0 = \|\overline{Q}\|_0$.

By induction on the equivalence, it is enough to show the result when Q is obtained form P by a Tietze transformation (T0) or (T1), which we do below. Throughout the section, we suppose fixed a 1-polygraph P. We first show that transformations (T0) are correct [13, tietze0-correct]:

Proposition 14.
Given a type X and a function $\partial : X \to \mathrm{U}\,P$, we have $\|\overline{P}\|_0 = \|\overline{P{\uparrow_0}\partial}\|_0$.

Proof.
By definition of $P{\uparrow_0}\partial$ and its geometric realization, we have a coequalizer

$$P_1 \sqcup X \underset{\partial^+ \sqcup \mathrm{id}_X}{\overset{\partial^- \sqcup \partial}{\rightrightarrows}} P_0 \sqcup X \longrightarrow \overline{P{\uparrow_0}\partial}$$

By computing the coequalizer partially on the left component of the coproduct, we find that $\overline{P{\uparrow_0}\partial}$ is the coequalizer of $X \underset{\mathrm{inr}}{\overset{\iota\circ\partial}{\rightrightarrows}} \overline{P} \sqcup X$. From which we easily deduce $\overline{P{\uparrow_0}\partial} = \overline{P}$. ◀ The transformations (T0) actually preserve, not only the presented set, but also the geometric realization. We now show that transformations (T1) are correct [13, tietze1-correct]:

Proposition 15.
Given a type X, a function $\partial : \mathrm{S}_0\,P \to \mathcal{U}$, and a family of functions $\partial_{x,y} : \partial(x,y) \to (x \overset{*}{\to} y)$ indexed by 0-spheres $(x,y) : \mathrm{S}_0\,P$, we have $\|\overline{P}\|_0 = \|\overline{P{\uparrow_1}\partial}\|_0$.

Proof.
By using the induction principle of geometric realization (Proposition 3), we can define a map $s : \overline{P} \to \overline{P{\uparrow_1}\partial}$ by

$s(\iota(x)) := \iota(x)$ $\mathrm{ap}_s(\gamma(a)) := \gamma(\mathrm{inl}(a))$ for $x : P_0$ and $a : x \to y$ in P_1. Conversely, we define a map $r : \overline{P{\uparrow_1}\partial} \to \overline{P}$ by induction by

$r(\iota(x)) := \iota(x)$ $\mathrm{ap}_r(\gamma(\mathrm{inl}(a))) := \gamma(a)$ $\mathrm{ap}_r(\gamma(\mathrm{inr}(b))) := \gamma^*_{x,y}(\partial_{x,y}(b))$ for $x : P_0$, $a : x \to y$ in P_1, and $b : \partial(x,y)$ for some $(x,y) : \mathrm{S}_0\,P$. By induction, they can be shown to form a section-retraction pair, which induces an equivalence on the 0-truncations. ◀

2.3 Completeness

We now show a converse completeness result: any two 1-polygraphs presenting the same set are Tietze equivalent. Throughout the section, we thus suppose fixed two 1-polygraphs P and Q such that we have an equality $\|\overline{P}\|_0 = \|\overline{Q}\|_0$. We write $f : \|\overline{P}\|_0 \to \|\overline{Q}\|_0$ and $g : \|\overline{Q}\|_0 \to \|\overline{P}\|_0$ for the components of the induced equivalence between the two types. Our goal is to show that the 1-polygraphs P and Q are equivalent. We should point out that this part has not (yet) been fully formalized in Agda.

In this section, we also need to suppose that the axiom of choice is satisfied. It is known to be consistent with homotopy type theory [19, Lemma 3.8.2], and can be formulated as follows:

Axiom 18.
The axiom of choice assumes an inhabitant of the type

$$\Pi(x : A).\|B(x)\|_{-1} \to \|\Pi(x : A).B(x)\|_{-1}$$

for every $A : \mathcal{U}$ and $B : A \to \mathcal{U}$.

Informally, this axiom states that any family of merely inhabited types is itself merely inhabited.

Proving the completeness result in Theorem 22 below requires us to first show some technical lemmas. We begin by observing that the maps f and g can be lifted to maps between the underlying types of 0-generators.

Proposition 19.
There merely exist maps $\hat{f} : \mathrm{U}\,P \to \mathrm{U}\,Q$ and $\hat{g} : \mathrm{U}\,Q \to \mathrm{U}\,P$ such that $|\iota(\hat{f}(x))|_0 = f(|\iota(x)|_0)$ $|\iota(\hat{g}(y))|_0 = g(|\iota(y)|_0)$ for $x : \mathrm{U}\,P$ and $y : \mathrm{U}\,Q$.

Proof.
We define a map f' by

$$f' \quad := \quad \mathrm{U}\,P \xrightarrow{\iota} \overline{P} \xrightarrow{|-|_0} \|\overline{P}\|_0 \xrightarrow{f} \|\overline{Q}\|_0 \longrightarrow \|\,\mathrm{U}\,Q\|_{-1}$$

where the rightmost map is the one defined in Proposition 8. By the axiom of choice, we deduce the existence of a map

$$\hat{f} : \|\,\mathrm{U}\,P \to \mathrm{U}\,Q\|_{-1}$$

satisfying the required identity. The map $\hat{g}$ is constructed similarly. ◀ We then show that any 1-generator in P induces a 1-cell in Q between the image of its endpoints.

Lemma 20.
Given a 1-generator $a : x \to y$ in P, there merely exists a 1-cell $\hat{f}(x) \xrightarrow{*} \hat{f}(y)$ in Q, that we call $\hat{f}(a)$ in the following.

Proof.
By definition of the geometric realization, the 1-generator a induces an equality $\gamma(a) : \iota(x) = \iota(y)$ in $\overline{P}$, and thus, by functoriality of f and $|\cdot|_0$, an equality

$$\mathrm{ap}_f(|\gamma(a)|_0) : f(|\iota(x)|_0) = f(|\iota(y)|_0)$$

Since, by Proposition 19, we have $|\iota(\hat{f}(x))|_0 = f(|\iota(x)|_0)$ (and similarly for y), there is an equality

$$|\iota(\hat{f}(x))|_0 = |\iota(\hat{f}(y))|_0$$

and we deduce, by Lemma 7, the existence of a 1-cell $\hat{f}(x) \xrightarrow{*} \hat{f}(y)$ as desired. ◀ Furthermore, the maps $\hat{f}$ and $\hat{g}$ are mutually inverse modulo post-composition by $|\iota(\cdot)|_0$.

Lemma 21.
For every $x : \mathrm{U}\,P$, we have $|\iota(\hat{g}(\hat{f}(x)))|_0 = |\iota(x)|_0$ in $\|\overline{P}\|_0$ (and similarly, we have $|\iota(\hat{f}(\hat{g}(x)))|_0 = |\iota(y)|_0$ for $y : \mathrm{U}\,Q$).

Proof.
Given $x : \mathrm{U}\,P$, by Proposition 19 and the fact that f and g are mutually inverse maps, we have $|\iota(\hat{g}(\hat{f}(x)))|_0 = g(|\iota(\hat{f}(x))|_0) = g(f(|\iota(x)|_0)) = |\iota(x)|_0$. ◀ Finally, we can show the completeness theorem.

Theorem 22 (Completeness).
Consider two 1-polygraphs P and Q such that $\|\overline{P}\|_0 = \|\overline{Q}\|_0$. Then, assuming the axiom of choice, P and Q are merely Tietze-equivalent.

Proof.
Starting from the polygraph P, we are going to construct step-by-step a polygraph P^3 that contains both P and Q and is Tietze-equivalent to P (and thus to Q by symmetry).
1.

We add the 0-generators of Q. We write $P^1 := P{\uparrow_0}\hat{g}$ for the polygraph obtained from P by performing a transformation (T0) with $\hat{g} : \mathrm{U}\,Q \to \mathrm{U}\,P$. Its type of 0-generators is $\mathrm{U}\,P \sqcup \mathrm{U}\,Q$ and the 1-generators are those of P together with one 1-generator $\hat{g}(x) \to x$ for every $x : \mathrm{U}\,Q$.

2.

We add the relations from P to Q. Given a 0-generator $x : \mathrm{U}\,P$, we claim that there exists a 1-cell $\hat{f}(x) \overset{*}{\to} x$ in P^1. By previous step, we have a cell

$$a : \hat{g}(\hat{f}(x)) \to \hat{f}(x)$$

Moreover, by Lemma 21, we have $|\iota(\hat{g}(\hat{f}(x)))|_0 = |\iota(x)|_0$, so the type $\|\hat{g}(\hat{f}(x)) \overset{*}{\to} x\|_{-1}$ is inhabited by Lemma 7, and we can suppose that we actually have a 1-cell

$$u : \hat{g}(\hat{f}(x)) \overset{*}{\to} x$$

because we are eliminating to a proposition. Hence, we can define a 1-cell $w_x : \hat{f}(x) \overset{*}{\to} x$ by $w_x := a ::' u$ for every $x : \mathrm{U}\,P$. Finally, we consider the function

$\partial : \mathrm{U}\,P^1 \to \Sigma((x,y) : \mathrm{S}_0\,P^1).(x \overset{*}{\to} y)\ x \mapsto ((\hat{f}(x), x), w_x)$ and define $P^2 := P^1{\uparrow_1}\partial$ by a transformation (T1'), see Proposition 11.

3.

We add the 1-generators of Q. Given a 1-generator $b : x \to y$ in Q, we have 1-generators $a : g(x) \to x$ and $c : g(y) \to y$ in P^2 (by the first step), and we have the 1-cell $g(a) : g(x) \overset{*}{\to} g(y)$ by Lemma 20. We therefore have a 1-cell

$$a \cdot g(b) \cdot c : x \overset{*}{\to} y$$

in Q. This induces a function

$$\partial : \Sigma((x,y) : \mathrm{S}_0\,Q).Q(x,y) \to \Sigma((x,y) : \mathrm{S}_0\,P^2).(x \overset{*}{\to} y)$$

which, to $(x,y) : \mathrm{S}_0\,Q$ and $a : Q(x,y)$ associates the pair consisting of (x,y) (or rather its image under the canonical inclusion) and the 1-cell constructed above. Finally, we define $P^3 := P^2{\uparrow_1}\partial$ by a transformation (T1').

In the 1-polygraph P^3 constructed above, the 0-generators are $\mathrm{U}\,P^3 = \mathrm{U}\,P \sqcup \mathrm{U}\,Q$ and the 1-generators are (up to canonical inclusion): those of P, those of Q (by the third step), $\hat{g}(x) \to x$ for $x : \mathrm{U}\,Q$ (by the first step), $\hat{f}(x) \to x$ for $x : \mathrm{U}\,P$ (by the second step). By exchanging the roles of P and Q, we see that Q is also Tietze-equivalent to the above 1-polygraph (up to standard isomorphisms such as commutation of coproducts). By transitivity, we conclude that P and Q are Tietze-equivalent. ◀

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